

FLUID AND ELECTROLYTE THERAPY FOR INFANTS AND CHILDREN

A. Introduction

1. The goals of intravenous (IV) fluid therapy are to prevent:
 - a. Dehydration.
 - b. Deficiencies of sodium and potassium.
 - c. Ketoacidosis.
 - d. Protein degradation.
2. There is no “one-size-fits all” approach to IV fluid therapy, as the prescription needs to be individualized to take into account the following:
 - a. Medical condition of the child.
 - b. Initial fluid, electrolyte and acid-base status.
 - c. Therapeutic goal.
 - d. Changes in the fluid, electrolyte and acid-base status with ongoing IV fluid therapy and losses.

B. Fluid Requirements for Children with Normal Hydration

I. *Daily maintenance fluid requirements via IV route:*

1. Association between metabolic rate and fluid requirement:
 - a. Fluid maintenance volume is the volume of daily fluid intake which replaces the insensible losses (from skin, lungs and stools) and the fluid required to excrete excess solute load (urea, creatinine, electrolytes) in a child with normal kidneys.
 - b. In a child with average normal kidney response: **1 ml of water is required for each kilocalorie consumed:**

	Daily fluid volume (ml per 100 kcal)
Insensible loss (IL)	50
Obligatory urine volume required to excrete solute load (renal solute load) (RSL)	66.7
Water of oxidation (WO)	16.7
Total fluid requirement = IL + RSL - WO	100

2. Caloric requirements:

Body weight (kg)	Daily caloric requirements (kcal/day)
3-10	100 kcal/kg
10-20	1000 + 50 kcal/kg for each kg >10kg
>20	1500 + 20 kcal/kg for each kg >20kg

Note: $IL - WO = 30 \text{ ml}/100 \text{ kcal}$ or $400 \text{ ml}/\text{m}^2$.

3. Maintenance fluid requirements for a well child with normal hydration but no oral intake (e.g. prior to elective surgery):

- a. Calculated using **Holliday-Segar method** which estimates caloric expenditure from weight alone:

Body Weight (kg)	Daily Maintenance Fluid
3-10	100 ml/kg
10-20	1000 ml + 50 ml/kg for each kg >10kg
>20	1500 ml + 20 ml/kg for each kg >20kg
Normal maximum amount	2500 ml per day

- b. This method is **NOT**:

- Suitable for babies <28 days old.
- For calculating oral maintenance fluid requirements.
- Used for patients who are overweight.

4. Fluid requirement of an overweight patient is based on age and **ideal body weight (IBW) (kg)**:

a. Weight for 50% percentile for patient's age
OR
b. Formula to get IBW (in adolescent and adults)
Male: $50 + 2.3 * (\text{height in cm} / 2.5 - 60)$
Female: $45.5 + 2.3 * (\text{height in cm} / 2.5 - 60)$

5. Maintenance fluid requirements for term neonates (<28 days old):

Age	Daily fluid requirements (ml/kg)
Day 1	50-60
Day 2	70-80
Day 3	80-100
Day 4	100-120
Day 5-28	120-150

6. Adjustment of maintenance fluid requirements may be needed in the following:

- a. Factors affecting insensible water loss:

Increased losses	% change	Decreased losses	% change
Prematurity	100-300	Enclosed incubator	25-50
Radiant warmer	50-100	Humidified air	15-30
Phototherapy	25-50	Sedation	5-25
Hyperventilation	20-30	Decreased activity	5-25
Increased activity	5-25	Hypothermia	5-15
Hyperthermia	12% per °C		

Example: A sick child on mechanical ventilation with humidified gases may need 50% less fluid than a well child (due to 25% decreased losses from inactivity and 25% decreased losses from humidified air).

- b. Maintenance fluid requirements may be reduced to 50-80% in inappropriate anti-diuretic hormone (ADH) secretion:
 - i. Post-surgery.
 - ii. Stress, pain, nausea or vomiting.
 - iii. Central nervous system disease.
 - iv. Liver disease.
 - v. Nephrotic syndrome.
 - vi. Pulmonary disease.
 - vii. Hypothyroidism.
 - viii. Hypoadrenalism.
 - ix. Medications e.g. carbamazepine, cyclophosphamide, vincristine, selective serotonin reuptake inhibitors.
- c. Underlying medical conditions with specific fluid requirements e.g. kidney failure and cardiac failure.

II. Daily electrolyte requirements:

Sodium (Na)	3-4 mmol/kg
Potassium (K)	2-3 mmol/kg
Chloride (Cl)	2-4 mmol/kg
Calcium (Ca)	0.5-1.0 mmol/kg (maximum 2 g/day in infants, and 3-4 g/day in older children)
Magnesium (Mg)	0.25 mmol/kg
Phosphate (PO ₄)	2 mmol/kg

III. Types of solutions for IV maintenance fluids:

1. Considerations:
 - a. Increasing reports of iatrogenic hyponatremia with catastrophic neurological outcomes seen in hospitalized children receiving hypotonic maintenance fluid therapy.
 - b. Cause of hyponatremia is due to a combination of a positive balance of electrolyte-free water (from the administration of hypotonic fluid) and the impaired ability to excrete hypotonic urine secondary to ADH secretion in sick patients.
 - c. Data supporting isotonic 0.9% saline solution as maintenance fluid therapy included several meta-analyses based on small clinical trials.
 - i. American Academy of Pediatrics (AAP) recommended that patients 28 days to 18 years of age requiring maintenance IV fluids should receive isotonic solutions because they significantly decrease the risk of developing hyponatremia.
 - ii. NICE 2015 guidelines recommended the initial use of isotonic crystalloids that contain sodium in the range of 131-154 mmol/L as maintenance fluid.
 - d. Physiologically, IV fluids, according to their tonicity, redistribute mainly to the interstitial fluid (isotonic fluids) or the interstitial and intracellular fluid compartment (hypotonic fluids).

- i. Isotonic fluids are typically employed in fluid resuscitation to correct an acute intravascular fluid deficit.
 - ii. Administration of hypotonic fluids generates the necessary osmotic driving force needed to allow movement of water from the extracellular to intracellular compartment, hydrating both compartments which is the goal of maintenance fluid therapy in hemodynamically stable fasting children.
- e. There is increasing evidence that administration of Cl-rich IV fluids (e.g. 0.9% saline which contains Cl exceeding that of plasma by approximately 60%), is associated with:
 - i. Increased risk of mortality.
 - ii. Direct effect on intrarenal perfusion, vasoconstriction, and glomerular filtration.
 - iii. Impaired coagulation (Cl-rich resuscitation fluid).
 - iv. Risk of fluid overload compared to a balanced crystalloid solutions (e.g. PlasmaLyte with Cl concentration closer to plasma Cl concentration).
 - v. Hypertension.

Note: Children with impaired kidney function are particularly at risk of above sequelae due to reduced ability to excrete excess Cl.
- f. 0.9% saline solution delivers Na and Cl load 2x more than 0.45% saline solution ml for ml, and can potentially result in Na and Cl overload in children.
- 2. Choice of maintenance fluids:
 - a. Children <40 kg: use 0.45% saline solution (hypotonic solution) as initial maintenance fluid therapy for well children and adjusted in response to monitoring during intravenous fluid therapy.
 - b. Children ≥40 kg: use 0.9% saline solution as 0.45% saline delivers a Na load of <4 mmol/kg/day, except for children who cannot tolerate Na or Cl overload.
 - c. Recommended fluid to be infused as maintenance therapy for well children with normal hydration is:

Children <40 kg: 0.45% saline with 5% dextrose
Children ≥40 kg: 0.9% saline with 5% dextrose
Add 1-2 mmol/kg/day of KCl (if there is no hyperkalemia or kidney impairment)

- d. Do **NOT** use 0.45% saline with 5% dextrose solution in the following situations:
 - i. Serum Na is low (<130 mmol/L).
 - ii. Initial treatment of acute neurological conditions e.g. meningitis and head injury.
 - iii. Conditions associated with syndrome of inappropriate antidiuretic hormone secretion (SIADH).
 - iv. Volume resuscitation.

e. Neonates:

- i. For term neonates ≤ 7 days old, 0.45% saline with 5% dextrose solution can provide excessive Na and inadequate glucose.
 - Consider a combination drip consisting of dextrose and added 20% sodium chloride (NaCl) to provide 4 mmol/kg/day of Na and a glucose delivery rate of 4-6 mcg/kg/min.
- ii. For term neonates ≥ 8 days old, 0.45% saline with 5% to 10% dextrose solution can be used safely.

IV. Daily maintenance fluid requirements via ORAL route:

1. Due to the low caloric density of milk (67 kcal/ml) and significant water loss in the feces with oral feeding, the recommended oral maintenance fluid volume is higher than intravenous requirement.

Age		Recommended daily oral fluid
1 st day		60 ml/kg/day
2 nd day		90 ml/kg/day
3 rd day		120 ml/kg/day
4 th day to 6 months		150 ml/kg/day
7 months to 1 year		100-120 ml/kg/day
1 to 3 years		900 ml/day
4 to 8 years		1200 ml/day
9 to 13 years	Boys	1800 ml/day
	Girls	1600 ml/day
14 to 18 years	Boys	2600 ml/day
	Girls	1800 ml/day

2. Types of oral fluids:
 - a. Age ≤ 6 months old: Mainly milk.
 - b. Age > 6 months old: Water, milk and other fluids once solid feeds are established

C. Fluid Requirements for Children with Dehydration

The fluid requirements for a dehydrated child should include ▪ Maintenance requirements + ▪ Deficit + ▪ Ongoing losses

I. Estimation of severity of dehydration:

Clinical Signs	Magnitude of Dehydration (Body weight loss)		
	Mild (3-5%)	Moderate (6-9%)	Severe (≥10%)
Behavior	Normal	Irritable or lethargic	Hyperirritable, drowsy or comatose
Breathing	Normal	Increased	Increased, deep
Thirst	Slight	Moderate	Intense
Mucous membrane	Dry	Very dry	Parched
Tears	Present	+/-	Absent
Skin turgor	Normal	Tenting	Severe tenting
Skin colour and temperature	Normal	Cool	Cool and mottled
Capillary refill	<2 sec	2-3 sec	>3 sec
Anterior fontanelle	Normal	+/-	Sunken
Eyes	Normal	Slightly sunken	Sunken
Urine output	Decreased	Oliguric	Azotemic
Blood pressure (BP)	Normal	Normal or postural drop	Decreased
Heart rate	±Increased	Increased	Marked increase

II. Initial resuscitation (in established or impending shock):

1. Initial resuscitation should be **fluid bolus of 20 ml/kg of 0.9% saline, Ringer's lactate or PlasmaLyte solution.**
 - Hypotensive: Give over 10 minutes.
 - Compensated shock: Give over 20-30 minutes.

Note: Patients should **not** have signs of fluid overload.
2. Limit crystalloid fluid boluses in specific pre-existing conditions (e.g. cardiac or kidney disease), diabetic ketoacidosis or trauma.
 - In trauma, give blood products early (bolus 10 ml/kg), preferentially over crystalloids.
3. The child should be reassessed (heart rate, perfusion, blood pressure, urine output) immediately after the fluid bolus and additional fluid bolus may be needed.
 - Seek expert advice early (e.g. from intensivists) if >40-60 ml/kg of IV bolus fluids are needed and/or inotropes are required.

III. Maintenance volume requirement:

1. Most children will tolerate standard fluid requirements as recommended by Holliday-Segar method above.
2. Some acutely ill children with increased ADH secretion may benefit from their maintenance fluid requirement being restricted to 50-80% of the normal recommended volume.

IV. **Deficit volume replacement:**

1. Fluid deficit in ml can be calculated following an estimation of the degree of dehydration expressed as % of body weight.

$$\text{Volume of fluid deficit (ml)} = \% \text{ dehydration} / 100 \times \text{weight (kg)} \times 1000$$

2. Recommended fluid to replace volume deficit:

For <40 kg and without hyponatremia: 0.45% saline with 5% dextrose
For ≥40 kg: 0.9% saline with 5% dextrose

3. Rate of replacement of fluid deficit depends on the child's illness:
 - a. Replacement may be rapid in most cases of gastroenteritis.
 - Give first half of the total volume (fluid deficit + maintenance - initial resuscitation volume) over first 8 hours followed by half of the total volume over the next 16 hours.
 - b. Replacement has to be slower in diabetic ketoacidosis and meningitis.
 - Replace total volume over 48 hours.

V. **Ongoing losses**

1. Ongoing losses (e.g. from diarrhea, nasogastric suction and drains) should be measured and replaced regularly if above combined fluid deficit/maintenance approach fails to correct the dehydration.
2. Replacement may be based on each previous hour, or each 4-hour period, depending on the situation.
3. Type of fluid used to replace ongoing losses depends on the electrolyte composition of the fluid being lost.
 - For example, nasogastric suction losses should be replaced with IV 0.45% saline with KCl 10 mmol added to each 500 ml of 0.45% saline

Body fluid	Na mmol/L	K mmol/L	Cl mmol/L	HCO ₃ mmol/L
Gastric juice	20-80	5-30	100-140	0
Pancreas	120-140	5-15	90-120	110
Bile	130-160	5-15	80-120	40
Small intestine	100-140	5-25	90-135	30
Ileostomy	45-135	3-15	20-115	30-110
Diarrhea	10-90	10-80	10-110	15-50
Sweat normal	10-30	3-10	10-35	0
Sweat cystic fibrosis	50-130	5-25	50-110	0
Burn exudate	140	5	110	20

VI. **Example of fluid therapy for severe isonatremic dehydration:**

A 10 month old boy with body weight of 10 kg was admitted for gastroenteritis with initial hypotension due to 15% dehydration. Laboratory results revealed Na 138 mmol/L, K 3 mmol/L and Cl 95 mmol/L.

- a. Initial management of hypovolemic shock:
0.9% saline 20 ml/kg (200 ml) was given over 10 mins. BP returned to normal after this.
- b. Subsequent fluid regime:
Maintenance fluid (ml) = $100 \text{ ml/kg} \times 10$
= 1000 ml
Fluid deficit (ml) = $\% \text{ dehydration}/100 \times \text{weight (kg)} \times 1000$
= 1500 ml
Remaining fluid deficit volume after fluid bolus = $1500 - 200 = 1300 \text{ ml}$
Total volume to give = Remaining fluid deficit + maintenance volume
= $1300 + 1000 = 2300 \text{ ml}$
Give first $\frac{1}{2}$ of 2300ml over 1st 8 hours and the rest over next 16 hours.

Fluid orders:

IV 5% dextrose + 0.45% saline 1150 ml over 1st 8 hours,

Followed by IV 5% dextrose + 0.45% saline 1150 ml over next 16 hours.

Add IV 7.45% KCl 2 mmol/kg/day to the 5% dextrose + 0.45% saline solution if there are no contraindications.

D. **Rehydration of Children with Severe Acute Malnutrition**

1. Severe acute malnutrition (SAM) is defined as mid-upper arm circumference <11.5 cm or weight-for-height Z-score <-3.
2. Rehydration of dehydrated children with SAM should be done orally rather than intravenously, when possible, to reduce the risk of overhydration and heart failure.
3. Rehydrate initially using standard WHO low-osmolality oral rehydration salt (ORS) solution (glucose 125 mmol/L, Na 45 mmol/L, K 40 mmol/L, Cl 70 mmol/L, osmolality 300 mmol/L).
 - a. Give at 5 ml/kg every 30 minutes for the first 2 hours.
 - b. If child is still dehydrated, give F-75 (starter formula with 75 kcal and 0.9 g protein per 100 ml) 5-10 ml/kg/hour every alternate hour for a maximum of 10 hours.
 - c. Assess frequently: Every 30 minutes for the first 2 hours and hourly thereafter.
4. If in shock, give IV resuscitation fluid at 15 ml/kg/hour using any of the following:
 - a. Ringer's lactate solution with 5% dextrose.
 - b. 5% dextrose + 0.45% saline solution with KCl 20 mmol/L.
 - c. Monitor every 5-10 min for signs of overhydration and congestive heart failure.
 - If signs develop, stop IV immediately.
5. Monitor serum electrolytes at baseline and 6 hourly until the child is fully hydrated

E. **Correction of Electrolyte and Acid-Base Abnormalities**

I. Hyponatremia:

1. Management of hyponatremia must take into consideration the patient's fluid status:
 - a. Fluid restriction in SIADH.
 - b. Diuretics in congestive heart failure, nephrotic syndrome, liver cirrhosis.
 - c. Na and water deficit replacement in hypovolemic hyponatremia.

2. In acute hyponatremia (<48 hours), Na correction can be done quickly.
3. **Recommended target rate of Na correction in chronic hyponatremia (> 48 hours): <8-10 mmol/L per day.**
 - Most reported cases of osmotic demyelination occurred after rates of correction that exceeded 12 mmol/L per day, although isolated cases occurred after only 9-10 mmol/L per day.
4. Na equivalent in fluid preparations:

Normal saline	= 154 mmol/L Na
D/S(R) or 0.45% saline	= 77 mmol/L Na
D/S(M) or 0.23% saline	= 38.5 mmol/L Na
3% saline	= 513 mmol/L Na

Note: Saline concentration should not exceed 3% in IV infusion.
5. Na equivalent in oral preparations:

NaCl tablet	= 8.5 mmol Na + 8.5 mmol Cl
20% NaCl solution	= 3.42 mmol Na/ml
Sodium bicarbonate (NaHCO ₃) tablet 500 mg	= 137 mg Na
	= 6 mmol Na + 6 mmol HCO ₃

Shohl's solution (Na citrate) = 1 mmol Na + 1 mmol citrate (= 3 mmol HCO₃) per ml

II. Hyponatremic dehydration:

Na Deficit (mmol) = 0.6 x body wt (kg) x (135 - serum [Na])
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1. Example of fluid therapy of asymptomatic hyponatremic dehydration with serum [Na] >120 mmol/L:

A 10 month old boy with body weight of 10 kg was admitted for vomiting and diarrhea with 10% dehydration. Laboratory results revealed Na 125 mmol/L, K 3.0 mmol/L and Cl 94 mmol/L.

- a. Calculate the total fluid volume required.

Fluid deficit (ml) = % dehydration/100 X weight (kg) X 1000 = 1000 ml

Maintenance fluid (ml) = 100 ml/kg x 10 = 1000 ml

Total fluid volume = fluid deficit + maintenance = 2000 ml

- b. Calculate the estimated Na load required to correct the hyponatremia.

- Maintenance Na = 4 mmol/kg/day = 4 x 10 = 40 mmol

- Na deficit associated with isonatremic volume loss (mmol)

= 140 mmol/L x fluid deficit volume (L) = 140 mmol

- Na deficit on top of isonatremic volume loss (mmol)

= 0.6 x weight (kg) x (135 - serum Na) = 0.6 x 10 x (135-125) = 60 mmol

- Total Na required = 40 + 140 + 60 = 240 mmol

- c. Giving 2 L of 0.45% saline would give Na load of 154 mmol which is insufficient to correct the hyponatremia.

- Hence, in this case to ensure adequate Na load to correct hyponatraemia, 2 L of **0.9% saline will be used** and this would give a Na load of 308 mmol.

- d. Using **sequential deficit/maintenance approach**:

- Give the fluid deficit volume over 1st 8 hours followed by maintenance volume in the next 16 hours.

Fluid orders:

IV 5% dextrose + 0.9% saline 1000 ml over 1st 8 hours.

Followed by IV 5% dextrose + 0.9% saline 1000 ml over next 16 hours.

Add IV 7.45% KCl 2 mmol/kg/day to the 5% dextrose + 0.9% saline if there is urine output.

2. Adrogué-Madias equation:

- a. Can be used to predict the change in serum Na concentration [Na] (mmol/L) following a therapeutic maneuver to avoid overly rapid correction of hyponatremia:

$$\Delta [\text{Na}] = \frac{\text{Infusate } [\text{Na}] - \text{serum } [\text{Na}]}{[0.6 \times \text{body wt (kg)}] + V} \times V$$

where $\Delta [\text{Na}]$ = change in serum [Na], V = infusate volume (L)

b. Example of use of Adrogué-Madias equation:

A 10 month old boy with body weight of 10 kg was admitted for vomiting and diarrhea with 10% dehydration. Laboratory results revealed Na 125 mmol/L, K 3.0 mmol/L and Cl 94 mmol/L.

Fluid orders:

IV 5% dextrose + 0.9% saline 1000 ml over 1st 8 hours.

Followed by IV 5% dextrose + 0.9% saline 1000 ml over next 16 hours.

Add IV 7.45% KCl 2 mmol/kg/day to the 5% dextrose + 0.9% saline solution if there is urine output.

Using the Adrogué-Madias equation:

- In the 1st 8 hours, change in serum [Na] will be from 125 mmol/L to 129 mmol/L (increase [Na] at 0.5 mmol/L per hour).
- In the next 16 hours, change in serum [Na] will be from 129 mmol/L to 133 mmol/L (increase [Na] at 0.25 mmol/L per hour).

- c. An improved Adrogué-Madias equation has recently been derived to determine volume of any infusate (V) to correct hyponatremia to target [Na]:

$$V (\text{L}) = \frac{\Delta [\text{Na}] \times [0.6 \times \text{body wt (kg)}]}{\text{Infusate } [\text{Na}] - \text{target } [\text{Na}]}$$

3. Management of symptomatic acute hyponatremic dehydration with serum [Na] <120 mmol/L:

- Hyponatremia is seldom symptomatic unless the serum [Na] is <120 mmol/L and if hyponatremia occurs quickly.
- In a child with hyponatremic seizures, give 3-5 ml/kg of 3% saline over 10-15 minutes.
 - Monitor serum [Na] and if seizures are ongoing, infusion should be repeated.
- To prevent further hyponatremic seizures, aim for serum [Na] target of 120-125 mmol/L.
 - Calculate the Na deficit and give the balance of the 3% saline infusion **over the next 60 minutes**.
- When serum [Na] of 120 mmol/L is achieved, embark on further correction of fluid deficit and Na deficit gradually **over the next 48 hours**.

- Include maintenance Na requirement AND
 - Correction of Na deficit associated with isonatremic volume loss (mmol)
= 140 mmol/L x fluid deficit volume (L) AND
 - Correction of Na deficit on top of isonatremic volume loss (mmol)
= 0.6 x weight (kg) x (target [Na] - serum [Na])
- e. Monitor the serum Na and fluid status every 4-6 hours and adjust the rehydration drip as needed to achieve correction of fluid deficit and gradual correction of hyponatremia.
- f. Example of fluid therapy for symptomatic acute hyponatremia:

A 10-month-old boy with body weight of 10 kg was admitted for vomiting and diarrhoea with 10% dehydration. He is having a generalized tonic-clonic seizure. Laboratory results revealed Na 110 mmol/L, K 3.5 mmol/L and Cl 95 mmol/L.

- a. Emergency management of hyponatremic seizure:
Give 3-5 ml/kg of 3% saline given over 10-15 minutes.
*In this case, **50 ml of 3% saline was given over 15 mins** with resolution of seizure.*
- b. Aim for target serum [Na] of 120-125 mmol/L to prevent further seizures:
Na deficit (mmol) = 0.6 x weight (kg) x (120-serum [Na]) = 60 mmol
This is equal to 117 ml of 3% saline.
*50 ml of 3% saline had been given earlier to stop the fits.
Hence another 67 ml of 3% saline can be given **over next 60 minutes**.*
- c. Further correction of serum [Na] over the next 48 hours after 120 mmol/L has been achieved:
- Determine Na load in IV therapy to correct serum Na from 120 to 125 mmol/L over next 24 hours.
 - Maintenance Na requirement = 4 mmol/kg/day = 4 x 10 = 40 mmol
 - Correct Na deficit associated with isonatremic volume loss (mmol)
= 140 mmol/L x fluid deficit volume (L) = 140 mmol
 - Correct Na deficit on top of isonatremic volume loss (mmol)
= 0.6 x weight (kg) x (125 - serum Na)
= 0.6 x 10 x (125 - 120) = 30 mmol
 - Total Na required in hyponatremic dehydration = 40 + 140 + 30 = 210 mmol
 - Amount of IV 5% dextrose + 0.9% saline to deliver 210 mmol Na in next 24 hours = 1370 ml
 - With 117 ml of 3% NaCl given earlier to correct severe hyponatraemia, total fluid received in 24 hours will be 1487 ml.
 - This volume can correct 50% fluid deficit volume and covers daily maintenance fluid volume in next 24 hours.

Fluid orders using combined deficit/maintenance approach:

IV 5% dextrose + 0.9% saline 1370 ml over the next 24 hours.

4. In hyponatremic patients with **severe symptoms (obtundation, coma)**, the **initial rate of correction can be 1-2 mmol/L per hour for several hours**.
- This limit can be cautiously exceeded if severe symptoms are not responding to correction according to the specified target.
5. Recommended indications for stopping the rapid correction of symptomatic hyponatremia are:
- Cessation of life-threatening manifestations.
 - Moderation of other symptoms.
 - Achievement of serum [Na] of 125-130 mmol/L, or even lower i.e. 120 mmol/L, if the baseline serum [Na] was <100 mmol/L.

III. Asymptomatic hyponatremia (<135 mmol/L) developing on IV fluid therapy:

1. If a child is receiving a hypotonic solution, change to an isotonic fluid.
2. Restrict maintenance IV fluids in children who are hypervolemic or at risk of hypervolemia by either:
 - a. Restricting maintenance fluids to 50-80% of routine maintenance needs, OR
 - b. Reducing maintenance fluids, calculated on the basis of insensible losses within the range of 300-400 ml/m²/day plus urinary output.

IV. Hypernatremic dehydration:

Free Water Deficit (L) = 0.6 x body wt (kg) x {1 – (serum [Na]/140)}

1. Fluid resuscitation if hypovolemic:
 - a. If patient is in shock, give 10-20 ml/kg of 0.9% saline over 20-30 minutes.
 - b. If oliguric, give 10-20 ml/kg of 0.45% saline hourly till urine flow is established.
2. Correction of free water deficit should be done over at least 48 hours, lowering the serum [Na] by not more than 10 mmol/L per day.
 - a. Maintenance fluid requirements should be added to the calculation.
 - b. Do not correct too rapidly, as this will result in cerebral edema, due to the presence of intracellular idiogenic osmoles.
 - c. Choose solutions that are not too relatively hypotonic to the current serum [Na].
 - Never use pure dextrose solution as IV correction.
 - d. Monitor serum [Na] every 4-6 hourly.
 - If hypernatremia worsens or is unchanged in 4-6 hours, review the fluid and consider changing to a fluid with more free water content.
3. Correct for hypocalcemia (seen in 10%).
4. Serum [Na] >180 mmol/L is an indication for peritoneal dialysis, using a dialysate fluid containing 150 mmol/L [Na].
5. Example of fluid therapy in hypernatremic dehydration:

A 6-month-old boy was admitted for fever, vomiting and poor feeding. He was irritable with sunken eyes. Capillary refill was 2 seconds. His weight was 5 kg (< 3rd percentile). Laboratory results revealed Na 170 mmol/L, K 4 mmol/L and Cl 125 mmol/L.

- a. Calculate the free water deficit:

$$\begin{aligned}\text{Free water deficit (L)} &= 0.6 \times \text{weight (kg)} \times [1 - (\text{serum [Na]}/140)] \\ &= 0.6 \times 5 \times [1 - (170/140)] = 0.64 \text{ L}\end{aligned}$$

*To bring down his serum [Na] from 170 mmol/L to 140 mmol/L by 10 mmol/L/day, we aim to give this **free water deficit volume over 3 days.***

- b. Maintenance volume = 100 ml/kg x 5 = 500 ml per day
- c. Volume of fluid per day = maintenance + 1/3 of free water deficit
= 500 + 213 = 713 ml
- d. Decide on type of fluid for correction:
Can start with 5% dextrose + 0.45% saline.

Fluid orders:

IV 5% dextrose + 0.45% saline 713 ml over 24 hours

After 4 hours, the repeat serum [Na] was 172 mmol/L and there was polyuria past 4 hours.

Use fluid of increased free water content instead: change to IV 5% dextrose + 0.23% saline.

Consider adding replacement urine losses ml for ml.

V. Potassium (K) deficit:

$\text{K Deficit (mmol)} \approx 0.6 \times \text{body wt (kg)} \times (4 - \text{serum K})$
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1. For severe hypokalemia (serum [K] ≤ 2.5 mmol/L) with symptoms (cardiac arrhythmia, paralysis, severe muscle weakness), give K infusion at 0.2 mmol/kg/hour for 2-3 hours with cardiac monitoring and recheck serum [K] at the end.
 - a. **Maximum infusion rate in general ward is 0.2 mmol/kg/hour while maximum infusion rate in the intensive care unit is 0.4 mmol/kg/hour.**
 - b. Infusion of >40 mmol/L K solutions via peripheral vein may result in phlebitis, hence infusion via central line is preferred with ECG monitoring.
2. For serum [K] between 2.5-3.5 mmol/L, give enteral or slow IV K supplementation.
 - a. Use enteral route for K supplementation whenever possible due to risk of cardiac arrhythmia with rapid IV administration.
 - b. Oral KCl 0.5-1 mmol/kg/dose (maximum 20 mmol) and repeated at 4-6 hours
 - Age ≤ 5 years: Maximum 1 mmol/kg
 - Age >5 years: Maximum 0.5 mmol/kgOR
 - c. Oral KCl 0.5-1.0 mmol/kg/dose (maximum 10 mmol) hourly until serum [K] normalizes:
 - Check serum [K] after every 3 doses.
3. K replacement should only be given when urine output is established.
4. Contraindications to adding K additive into IV fluids if there is:
 - a. Presence of hyperkalemia (serum K >5 mmol/L)
 - b. At risk of hyperkalemia:
 - Use of potassium-sparing diuretics or renin-angiotensin system blockers.
 - Adrenal insufficiency.
 - Severe tissue damage e.g. burns or crush injury.
 - Tumor lysis syndrome.
 - c. Abnormal kidney function as suggested by elevated serum creatinine.
 - d. At risk of kidney impairment suggested by anuria for at least 12 hours or oliguria for at least 16 hours:
 - Acute glomerulonephritis.
 - Acute kidney injury.
5. K equivalent in oral preparations:
Span K tablet = 600 mg K = 8 mmol K
Mixture KCl = 500 mg K per 5 ml = 1.34 mmol K + 1.34 mmol Cl per ml
Mixture K citrate 30% = 2.8 mmol K + 0.9 mmol citrate per ml
K citrate tablet = 1080 mg = 10 mmol K + 3.3 mmol citrate

VI. Metabolic acidosis:

$$\begin{aligned} &\text{Bicarbonate (HCO}_3\text{) deficit (mmol)} \\ &= 0.3 \times (\text{Base Deficit}) \times \text{body wt (kg)} \\ &\text{OR} \\ &= 0.3 \times (\text{Desired [HCO}_3\text{]} - \text{Actual [HCO}_3\text{]}) \times \text{body wt (kg)} \end{aligned}$$

1. This formula accounts for half correction of the HCO₃ deficit.
2. Treatment of acute metabolic acidosis:
 - a. Consider giving IV NaHCO₃ especially if serum [HCO₃] <12 mmol/L.
 - In acidosis secondary to volume depletion, target a desired [HCO₃] of 15 mmol/L, as volume replacement is sufficient to correct the remaining acidosis.
 - In renal tubular acidosis, correct the acidosis to a desired [HCO₃] of 24 mmol/L.
 - In diabetic ketoacidosis, correct the acidosis with NaHCO₃ if pH is ≤7.0 targeting pH of at least 7.1.
 - b. Rate of correction depends on target [HCO₃] and clinical situation.
 - Give half of the HCO₃ deficit as bolus dose if pH is ≤7.0 followed by the other half as infusion over 2-4 hours.
 - c. Use 4.2% NaHCO₃ in neonates and infants.
3. Formulations:
 - 1 ml 8.4% NaHCO₃ = 1 mmol Na + 1 mmol HCO₃.
 - NaHCO₃ tablet 500 mg
= 137 mg Na = 6 mmol Na + 6 mmol HCO₃
 - Shohl's solution = 1 mmol Na + 1 mmol citrate (3 mmol HCO₃) per ml
4. Correct simultaneously K and Ca deficits during correction of metabolic acidosis.

F. Monitoring

1. Monitoring regime is essential for all children on IV hydration.
2. An accurate daily intake and output record should be kept.

Time	Parameters to be monitored
Prior to IV fluid administration	• Weight • Serum Na, K, Cl, HCO₃, urea, creatinine
Patient receiving replacement IV fluids for ongoing losses 24 hours later and subsequent days of IV therapy	• Daily weight • Daily serum Na, K (Note: More frequent monitoring 4-6 hourly if serum electrolytes are abnormal)
Patient still receiving IV maintenance fluids at >50% of calculated maintenance levels 24 hours later and subsequent days of IV therapy	• Daily weight • Periodic serum Na, K (Note: More frequent monitoring if serum electrolytes are abnormal)

D. Osmolality and Tonicity of Available Intravenous Solutions

IV solution	Glucose mmol/L	Na* mmol/L	Cl mmol/L	Lactate mmol/L	K mmol/L	Osmolality mOsm/L	Tonicity
0.9% saline	-	154	154	-	-	308	Isotonic
0.45% saline	-	77	77	-	-	154	Hypotonic
3% saline	-	513	513	-	-	1026	Hypertonic
5% dextrose	278	-	-	-	-	278	Very hypotonic
10% dextrose	555	-	-	-	-	555	Very hypotonic
0.9% saline + 5% dextrose	278	154	154	-	-	586	Isotonic
0.45% saline + 5% dextrose	278	77	77	-	-	432	Hypotonic
0.23% saline + 5% dextrose	278	38	38	-	-	355	Very hypotonic
Ringer's lactate	-	130	110	28	4	280	Almost isotonic
5% albumin	-	140	125	-	-	260	Isotonic
PlasmaLyte	23	140	98	Acetate: 27 mmol/L Gluconate: 23 mmol/L	5	295	Isotonic
Plasma (reference)		135-145	95-108		3.5-5.3	275-295	

*Solutions that contain [Na] in the range of 131-154 mmol/L are considered isotonic.

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This information aims to guide you in the management of children's diseases and reflects current practice in the Khoo Teck Puat-National University Children's Medical Institute. Considerable care has been taken to ensure that the protocols are accurate but they are not a substitute for asking advice from those more familiar with the problems patients present - if any aspect of patient care is unclear consult a more senior or subspecialist doctor.

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